

Online Appendix

From Income to Wealth Inequality: Trickle-Down vs. Capital Gains

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This appendix accompanies the paper “From Income to Wealth Inequality: Trickle-Down vs. Capital Gains.”

A Proofs of the Analytical Results

A.1 Additional details on the analytical model

The economy is composed of two agents who differ in their level of permanent income z_i . We index agents as 1 and 2 (1 being the rich, 2 the poor, so that $z_1 > z_2$). We also impose that $z_1 + z_2 = 1$, so that an increase in inequality is a mean-preserving spread. They have preferences towards consumption and wealth. Their maximization problem writes

$$\max_{c_t, a_t} V(a_{t-1}) = \frac{c_t^{1-\sigma}}{1-\sigma} + \frac{a_t^{1-\sigma_a}}{1-\sigma_a} + \beta V(a_t)$$

subject to $a_t + c_t = a_{t-1}(1 + r_t^a) + z_t w_t$. The first-order condition with respect to a_t is

$$c_t^{-\sigma} = a_t^{-\sigma_a} + \beta V'(a_t).$$

The envelope condition gives $V'(a_{t-1}) = (1 + r_t^a)c_t^{-\sigma}$, so that $V'(a_t) = (1 + r_{t+1}^a)c_{t+1}^{-\sigma}$ and, the Euler equation is

$$c_t^{-\sigma} = a_t^{-\sigma_a} + \beta(1 + r_{t+1}^a)c_{t+1}^{-\sigma}.$$

At the steady-state, $a_t = a_{t-1}$ so that $c = ar + wz$ and

$$1 - \beta(1 + r) = \frac{a^{-\sigma_a}}{c^{-\sigma}}.$$

A.2 Assumption 1 and the stability of the household steady state

Substituting the budget constraint into the Euler equation eliminates consumption and reduces the household problem to a single second-order difference equation in wealth,

$$\Psi(a_{t-1}, a_t, a_{t+1}) \equiv [Ra_{t-1} + wz - a_t]^{-\sigma} - a_t^{-\sigma_a} - \beta R[Ra_t + wz - a_{t+1}]^{-\sigma} = 0,$$

where $R \equiv 1 + r$ and $k \equiv 1 - \beta R > 0$. Setting $a_{t-1} = a_t = a_{t+1} = a$ recovers the steady-state Euler equation,

$$\Psi(a, a, a) = kc^{-\sigma} - a^{-\sigma_a} = -c^{-\sigma}F(a), \quad F(a, z, w, r) \equiv (ar + wz)^{\sigma}a^{-\sigma_a} + \beta(1 + r) - 1,$$

with $c = ra + wz$. Differentiating F gives $F_a = -a^{-\sigma_a-1}c^{\sigma-1}D$, where

$$D \equiv \sigma_a wz - (\sigma - \sigma_a)ar,$$

so that $F_a < 0$ if and only if $D > 0$, which is Assumption 1. Let $\Psi_{-1}, \Psi_0, \Psi_{+1}$ denote the partial derivatives of Ψ with respect to its three arguments, evaluated at a steady state a :

$$\begin{aligned}\Psi_{-1} &= -\sigma R c^{-\sigma-1} < 0, & \Psi_{+1} &= -\beta R \sigma c^{-\sigma-1} < 0, \\ \Psi_0 &= \sigma c^{-\sigma-1} (1 + \beta R^2) + \sigma_a a^{-\sigma_a-1} > 0.\end{aligned}$$

Linearising gives $\Psi_{+1}\hat{a}_{t+1} + \Psi_0\hat{a}_t + \Psi_{-1}\hat{a}_{t-1} = 0$, with normalised characteristic polynomial $P(\lambda) = \lambda^2 + (\Psi_0/\Psi_{+1})\lambda + \Psi_{-1}/\Psi_{+1}$. Two properties pin down the local dynamics.

(i) *Reciprocity*. For $\beta > 0$ the product of the roots is

$$\lambda_1 \lambda_2 = \frac{\Psi_{-1}}{\Psi_{+1}} = \frac{R}{\beta R} = \frac{1}{\beta} > 1,$$

so $\lambda_2 = 1/(\beta\lambda_1)$: the roots come in reciprocal pairs, the usual structure of a discounted variational problem. In particular they can never both lie inside the unit circle, so the steady state is either saddle-path stable or a source.

(ii) *The test at $\lambda = 1$* . Evaluating the polynomial at $\lambda = 1$ sums the three partial derivatives, which is the derivative of Ψ when its arguments move together — that is, along the locus of steady states. Since $\Psi(a, a, a) = -c^{-\sigma}F(a)$ and $F = 0$ at a steady state,

$$\Psi_{-1} + \Psi_0 + \Psi_{+1} = \frac{d}{da} \Psi(a, a, a) = -c^{-\sigma}F_a.$$

Dividing by $\Psi_{+1} < 0$ and using $c^\sigma a^{-\sigma_a} = k$,

$$(1 - \lambda_1)(1 - \lambda_2) = P(1) = \frac{c}{\beta R \sigma} F_a = -\frac{k}{\beta R \sigma a} D. \quad (1)$$

If $D > 0$, then $P(1) < 0$: exactly one root lies inside the unit circle and the steady state is saddle-path stable (the roots are then necessarily real, since complex conjugates would give $P(1) = |1 - \lambda_1|^2 > 0$). If $D < 0$, then $P(1) > 0$ and, by (i), both roots lie outside the unit circle: the steady state is a source and no path converges to it. Assumption 1 therefore selects the stable steady state. The equivalence in (1) is not a coincidence: $\lambda = 1$

describes a permanent constant deviation, that is, a move to a neighbouring steady state, which the steady-state equation admits only where it is degenerate, $F_a = 0$. The same condition governs stability and the applicability of the implicit function theorem.

Two remarks follow. First, since $\Psi(a, a, a) = -c^{-\sigma}F(a)$ and $\sigma > \sigma_a$, F has at most two zeros in a : generically either two — satisfying $D > 0$ and $D < 0$ respectively — or none. They collide when $ra/c = \sigma_a/\sigma$, beyond which the household has no finite steady-state wealth. Second, when $\beta = 0$, as in Propositions 3 and 4, the system reduces to the first-order map $a_t = g(a_{t-1})$ with $g'(a) = \sigma Ra/(\sigma a + \sigma_a c)$, and $g'(a) < 1$ is again equivalent to $D > 0$.

A.3 Proof of Proposition 1

Let us start from the asset-market clearing condition $A = a_1 + a_2 = q$. $A(z, w(r), r)$ is a function of z, r and $w(r)$, while the value of the firm is a function of the interest rate $q(r)$. Taking the log on both sides and totally differentiating the asset market equation yields

$$\begin{aligned} dr &= -\frac{\partial \log A}{\varepsilon^A + \varepsilon^q} \\ d \log(q/Y) &= \frac{\varepsilon^q - \varepsilon^Y}{\varepsilon^A + \varepsilon^q} \cdot \partial \log A, \end{aligned}$$

To show that $dr < 0$, we show that (i) $\partial \log A$ is positive, (ii) ε^q is positive, (iii) that ε^A is positive. To show that the wealth to output increases, we finally show that $\varepsilon^q > \varepsilon^Y$.

1. Positivity of $\partial \log A$

Now, let us prove that $\partial \log A > 0$. Recall from Online Appendix A.2 the steady-state Euler equation

$$F(a, z, w, r) \equiv (ar + wz)^\sigma a^{-\sigma_a} + \beta(1 + r) - 1 = 0.$$

By the Implicit Function Theorem, provided $F_a \neq 0$, we have

$$\frac{\partial a}{\partial z} = -\frac{F_z}{F_a}.$$

Taking the derivative, we obtain

$$\begin{aligned} F_a &= a^{-\sigma_a-1} c^{\sigma-1} [(\sigma - \sigma_a)ar - \sigma_a wz] \\ F_z &= \sigma w a^{-\sigma_a} c^{\sigma-1}. \end{aligned}$$

With $D \equiv \sigma_a wz - (\sigma - \sigma_a)ar$ as in Online Appendix A.2, we then have

$$\frac{\partial a}{\partial z} = \frac{\sigma w \cdot a}{D},$$

which is positive if $D > 0$. The second derivative can be written as

$$\frac{\partial^2 a}{\partial z^2} = \sigma w \left(\frac{\partial a}{\partial z} D^{-1} - a D^{-2} \frac{\partial D}{\partial z} \right)$$

with $\partial D / \partial z = \sigma_a w - (\sigma - \sigma_a)r \partial a / \partial z$. Thus,

$$\begin{aligned} \frac{\partial^2 a}{\partial z^2} &= \sigma w \left(\frac{\partial a}{\partial z} D^{-1} - a D^{-2} \left(\sigma_a w - (\sigma - \sigma_a)r \frac{\partial a}{\partial z} \right) \right) \\ &= \frac{\sigma w}{D} \left(\frac{\partial a}{\partial z} \left(1 + \frac{ar(\sigma - \sigma_a)}{D} \right) - \frac{a\sigma_a w}{D} \right) \\ &= \frac{\sigma w a}{D^2} \left(\sigma w \left(1 + \frac{ar(\sigma - \sigma_a)}{D} \right) - \sigma_a w \right) \\ &= \frac{\sigma w^2 a(\sigma - \sigma_a)}{D^2} \left(1 + \frac{\sigma ar}{D} \right) > 0. \end{aligned}$$

So $a(z; r, w)$ is increasing and convex in z . Since $A = a(z_1) + a(z_2)$ and $z_1 + z_2 = 1$, convexity implies that a mean-preserving spread ($dz_1 = -dz_2 > 0$) increases aggregate savings: $\partial \log A > 0$.

2. Positivity of ε^q

For convenience, define

$$\varepsilon^K \equiv -\frac{\partial \log K}{\partial r} = \frac{1}{(1 - \alpha)(r + \delta)}.$$

Recall that $q = K + V$ where $V \equiv \Pi / r$ with $\Pi \equiv \alpha^{\frac{\mu-1}{\mu}} Y$ denoting gross monopoly rents.

Since $\varepsilon^Y \equiv -\partial \log Y / \partial r = \alpha \varepsilon^K$, differentiating gives

$$-\frac{\partial q}{\partial r} = K \varepsilon^K + \frac{\Pi \varepsilon^Y + V}{r}.$$

Using $\Pi = rV$, we rewrite the second term as

$$\frac{\Pi \varepsilon^Y + V}{r} = V \left(\varepsilon^Y + \frac{1}{r} \right).$$

Now,

$$\varepsilon^Y + \frac{1}{r} = \frac{\alpha r + (1 - \alpha)(r + \delta)}{r(1 - \alpha)(r + \delta)} = \varepsilon^K \cdot \frac{r + (1 - \alpha)\delta}{r},$$

so that

$$-\frac{\partial q}{\partial r} = \varepsilon^K \left(q + V \frac{(1 - \alpha)\delta}{r} \right).$$

Dividing by q ,

$$\varepsilon^q = \varepsilon^K \left(1 + \frac{V}{q} \frac{(1 - \alpha)\delta}{r} \right) > 0, \quad (2)$$

since every factor is positive.

3. Positivity of ε^A

Now let us focus on ε^A . The aggregate semi-elasticity of asset demand can be decomposed as a wealth-weighted sum of individual semi-elasticities:

$$\varepsilon^A = \frac{d \log A}{dr} = \frac{1}{A} \sum_i \frac{da_i}{dr} = \frac{1}{A} \sum_i a_i \varepsilon^{a_i}. \quad (3)$$

Each individual semi-elasticity can in turn be decomposed into a direct effect and an indirect effect through the wage:

$$\varepsilon^{a_i} = \frac{d \log a_i(z, r, w(r))}{dr} = \frac{\partial \log a_i}{\partial r} + \frac{\partial \log a_i}{\partial \log w} \cdot \frac{d \log w}{dr} = \varepsilon^{a_i, r} + \varepsilon^{a_i, w} \varepsilon^{w, r}. \quad (4)$$

From the firm's first-order condition $(r + \delta) = \frac{\alpha}{\mu} ZK^{\alpha-1}$, we obtain

$$\varepsilon^{w,r} \equiv \frac{d \log w}{dr} = -\frac{\mu K}{w}. \quad (5)$$

Applying the Implicit Function Theorem to the household's Euler equation yields

$$\frac{\partial a_i}{\partial r} = \frac{\sigma a_i^2 + \beta a_i^{\sigma_a+1} c_i^{1-\sigma}}{D_i}, \quad \frac{\partial a_i}{\partial w} = \frac{\sigma z_i a_i}{D_i}, \quad (6)$$

where $D_i \equiv \sigma_a w z_i - (\sigma - \sigma_a) r a_i$. Converting to (semi-)elasticities,

$$\varepsilon^{a_i,r} = \frac{\partial \log a_i}{\partial r} = \frac{\sigma a_i + \beta a_i^{\sigma_a} c_i^{1-\sigma}}{D_i}, \quad \varepsilon^{a_i,w} = \frac{\partial \log a_i}{\partial \log w} = \frac{\sigma w z_i}{D_i}. \quad (7)$$

Substituting into the aggregate decomposition,

$$\varepsilon^A = \frac{1}{A} \sum_i a_i \left(\frac{\sigma a_i + \beta a_i^{\sigma_a} c_i^{1-\sigma}}{D_i} - \frac{\sigma w z_i}{D_i} \cdot \frac{\mu K}{w} \right) = \frac{\sigma}{A} \sum_i \frac{a_i}{D_i} (a_i - \mu K z_i) + \frac{\beta}{A} \sum_i \frac{a_i^{\sigma_a+1} c_i^{1-\sigma}}{D_i}. \quad (8)$$

The second sum is manifestly positive since $a_i^{\sigma_a+1} c_i^{1-\sigma} / D_i > 0$ for all i under our maintained assumption $D_i > 0$. For the first sum, we show that $\sum_i (a_i / D_i) (a_i - \mu K z_i) > 0$.

Using $a_2 - \mu K z_2 = (A - \mu K) - (a_1 - \mu K z_1)$, we write

$$\sum_i \frac{a_i}{D_i} (a_i - \mu K z_i) = \underbrace{(a_1 - \mu K z_1)}_{>0} \underbrace{\left(\frac{a_1}{D_1} - \frac{a_2}{D_2} \right)}_{>0} + \underbrace{\frac{a_2}{D_2}}_{>0} \underbrace{(A - \mu K)}_{\geq 0}, \quad (9)$$

a sum of two non-negative terms, the first strictly positive. We verify each sign in turn.

- (i) $a_1 - \mu K z_1 > 0$. We show that $a_1 / z_1 > a_2 / z_2$. This suffices because cross-multiplying yields $a_1 z_2 > a_2 z_1$, and substituting $z_2 = 1 - z_1$:

$$a_1(1 - z_1) > a_2 z_1 \iff a_1 > (a_1 + a_2) z_1 = A z_1.$$

Moreover, $A \geq \mu K$ since $A = q = K + \alpha \frac{\mu-1}{\mu} \frac{Y}{r}$ and $\alpha Y = \mu(r + \delta)K$ yield $A =$

$K(1 + \frac{(\mu-1)(r+\delta)}{r}) \geq \mu K$, so $a_1 > Az_1 \geq \mu Kz_1$.

We prove $a_1/z_1 > a_2/z_2$ by contradiction; the argument uses the steady-state Euler equation and therefore holds at the steady state only. Suppose $a_1/z_1 \leq a_2/z_2$, i.e. $a_1z_2 \leq a_2z_1$. From the budget constraint $c_i = a_i r + wz_i$,

$$c_1a_2 - c_2a_1 = w(a_2z_1 - a_1z_2) \geq 0,$$

so $c_1/c_2 \geq a_1/a_2$. Now, the Euler equation $c_i^\sigma a_i^{-\sigma_a} = 1 - \beta(1+r)$ takes the same value for both agents, so

$$\left(\frac{c_1}{c_2}\right)^\sigma = \left(\frac{a_1}{a_2}\right)^{\sigma_a} \implies \frac{c_1}{c_2} = \left(\frac{a_1}{a_2}\right)^{\sigma_a/\sigma}.$$

Both agents are on the stable branch of the steady-state equation (Assumption 1), along which $\partial a/\partial z = \sigma w a/D > 0$, so $z_1 > z_2$ implies $a_1 > a_2$ and $a_1/a_2 > 1$. Because $\sigma_a/\sigma < 1$ (the non-homotheticity assumption), concavity of $t \mapsto t^{\sigma_a/\sigma}$ implies $(a_1/a_2)^{\sigma_a/\sigma} < a_1/a_2$, so $c_1/c_2 < a_1/a_2$. This contradicts $c_1/c_2 \geq a_1/a_2$.

- (ii) $a_1/D_1 > a_2/D_2$. Cross-multiplying, $a_1D_2 - a_2D_1 = \sigma_a w(a_1z_2 - a_2z_1) > 0$, where the inequality follows from $a_1/z_1 > a_2/z_2$ established in (i).
- (iii) $A \geq \mu K$. This follows from $A = K(1 + (\mu-1)(r+\delta)/r) \geq \mu K$, with equality under perfect competition ($\mu = 1$).

4. $\varepsilon^q > \varepsilon^Y$

From the production function, $\varepsilon^Y \equiv -d \log(Y)/dr = \alpha \varepsilon^K$. Using our previous result, we have

$$\varepsilon^q - \varepsilon^Y = \varepsilon^K \left(1 + \frac{V(1-\alpha)\delta}{q r}\right) - \alpha \varepsilon^K = \varepsilon^K(1-\alpha) \left(1 + \frac{V\delta}{q r}\right) > 0.$$

5. The wage

Since $w = (1 - \alpha)Y$, $d \log w = d \log Y = -\varepsilon^Y dr = -\alpha \varepsilon^K dr > 0$, which is item 2 of the proposition.

A.4 Proof of Proposition 2

We show that $\varepsilon^{a_1} > \varepsilon^{a_2}$, which implies that $\varepsilon^{a_1} > \varepsilon^A$.

Preliminary: $D_i > 0$ for $i = 1, 2$.

Recall that $D_i \equiv \sigma_a w z_i - (\sigma - \sigma_a) r a_i$, so that $D_i > 0$ is equivalent to $r a_i / c_i < \sigma_a / \sigma$, which Assumption 1 imposes on both agents: each is at the saddle-path-stable root of its steady-state equation (Online Appendix A.2). Since $a_1 / z_1 > a_2 / z_2$ (step 3(i) of Proposition 1) and $r a_i / c_i$ increases with a_i / z_i , the condition binds for the rich agent.

From equation (8), the individual total semi-elasticity reads

$$\varepsilon^{a_i} = \frac{\sigma(a_i - \mu K z_i) + \beta a_i^{\sigma_a} c_i^{1-\sigma}}{D_i}. \quad (10)$$

Since $D_1, D_2 > 0$, the sign of $\varepsilon^{a_1} - \varepsilon^{a_2}$ is determined by the numerator

$$\mathcal{N} \equiv [\sigma(a_1 - \mu K z_1) + \beta a_1^{\sigma_a} c_1^{1-\sigma}] D_2 - [\sigma(a_2 - \mu K z_2) + \beta a_2^{\sigma_a} c_2^{1-\sigma}] D_1.$$

Separating the σ and β terms,

$$\mathcal{N} = \underbrace{\sigma[(a_1 - \mu K z_1) D_2 - (a_2 - \mu K z_2) D_1]}_{\mathcal{N}_1} + \underbrace{\beta[a_1^{\sigma_a} c_1^{1-\sigma} D_2 - a_2^{\sigma_a} c_2^{1-\sigma} D_1]}_{\mathcal{N}_2}.$$

First term. Expanding,

$$(a_1 - \mu K z_1) D_2 - (a_2 - \mu K z_2) D_1 = (a_1 D_2 - a_2 D_1) - \mu K (z_1 D_2 - z_2 D_1).$$

From step 3(ii) of Proposition 1, $a_1 D_2 - a_2 D_1 = \sigma_a w (a_1 z_2 - a_2 z_1)$. For the second cross-

product, substituting $D_i = \sigma_a w z_i - (\sigma - \sigma_a) a_i r$ gives

$$z_1 D_2 - z_2 D_1 = (\sigma - \sigma_a) r (a_1 z_2 - a_2 z_1).$$

Combining,

$$\mathcal{N}_1 = \sigma (a_1 z_2 - a_2 z_1) [\sigma_a w - \mu (\sigma - \sigma_a) r K].$$

Since $a_1/z_1 > \mu K$ (step 3(i): $a_1 > A z_1 \geq \mu K z_1$), the condition $D_1 > 0$ implies $\sigma_a w > (\sigma - \sigma_a) a_1 r / z_1 > \mu (\sigma - \sigma_a) K r$. Together with $a_1 z_2 - a_2 z_1 > 0$, this yields $\mathcal{N}_1 > 0$.

Second term. From the steady-state Euler equation $c_i^\sigma a_i^{-\sigma_a} = k$, where $k \equiv 1 - \beta(1 + r) > 0$ as in Online Appendix A.2, we obtain $a_i^{\sigma_a} = c_i^\sigma / k$ and hence $a_i^{\sigma_a} c_i^{1-\sigma} = c_i / k$. Therefore

$$\mathcal{N}_2 = \frac{\beta}{k} (c_1 D_2 - c_2 D_1).$$

Using the budget constraint $c_i = a_i r + w z_i$ together with the cross-products computed above,

$$\begin{aligned} c_1 D_2 - c_2 D_1 &= r(a_1 D_2 - a_2 D_1) + w(z_1 D_2 - z_2 D_1) \\ &= r \sigma_a w (a_1 z_2 - a_2 z_1) + w (\sigma - \sigma_a) r (a_1 z_2 - a_2 z_1) \\ &= \sigma w r (a_1 z_2 - a_2 z_1) > 0, \end{aligned}$$

so $\mathcal{N}_2 > 0$.

Since $\mathcal{N} = \mathcal{N}_1 + \mathcal{N}_2 > 0$, we conclude $\varepsilon^{a_1} > \varepsilon^{a_2}$. From the weighted-average decomposition $\varepsilon^A = (a_1/A) \varepsilon^{a_1} + (a_2/A) \varepsilon^{a_2}$, this implies $\varepsilon^{a_1} > \varepsilon^A$, i.e. $\varepsilon^{a_1} - \varepsilon^A > 0$. Since $dr < 0$, the trickle-down effect $(\varepsilon^{a_1} - \varepsilon^A) dr < 0$: the wealth of the rich declines proportionally more than aggregate wealth.

A.5 Proof of Proposition 3

We maintain $\beta = 0$ throughout.

Welfare derivative. With $\beta = 0$, per-period utility at the steady state is $U_i = c_i^{1-\sigma} / (1 - \sigma) + a_i^{1-\sigma_a} / (1 - \sigma_a)$. By the envelope theorem, the welfare effect of a marginal price change on agent i (at initial wealth a_i) is

$$\frac{dU_i}{dr} = c_i^{-\sigma} \left[a_i + z_i \frac{dw}{dr} \right].$$

From the production side, $w = (1 - \alpha)ZK^\alpha$ and $(r + \delta) = \frac{\alpha}{\mu} \frac{Y}{K}$, so that $dw/dr = -\alpha Y / (r + \delta) = -\mu K$. Using the household first-order condition $c_i^{-\sigma} = a_i^{-\sigma_a}$:

$$\frac{dU_i}{dr} = a_i^{-\sigma_a} (a_i - \mu K z_i).$$

Monotonicity of $\Phi(a) \equiv a^{-\sigma_a} (a - \mu K z(a))$. Substituting $z(a) = (a^\phi - ar)/w$ from the budget constraint ($c = a^\phi$ and $c = ar + wz$, with $\phi \equiv \sigma_a / \sigma$):

$$\Phi(a) = \left(1 + \frac{\mu K r}{w} \right) a^{1-\sigma_a} - \frac{\mu K}{w} a^{\phi-\sigma_a}.$$

The exponents are $1 - \sigma_a > 0$ (since $\sigma_a < 1$) and $\phi - \sigma_a = \sigma_a(1 - \sigma) / \sigma < 0$ (since $\sigma > 1$). Differentiating:

$$\Phi'(a) = \underbrace{(1 - \sigma_a) \left(1 + \frac{\mu K r}{w} \right) a^{-\sigma_a}}_{>0} + \underbrace{\frac{\mu K}{w} \cdot \frac{\sigma_a(\sigma - 1)}{\sigma} \cdot a^{\phi-\sigma_a-1}}_{>0} > 0,$$

since both terms are strictly positive. Therefore $a_1 > a_2$ implies $\Phi(a_1) > \Phi(a_2)$, i.e., $dU_1/dr > dU_2/dr$. Since $dr < 0$:

$$\Delta U_2 = \frac{dU_2}{dr} dr > \frac{dU_1}{dr} dr = \Delta U_1.$$

Full redistribution at $\mu = 1$. Under perfect competition, $V = 0$ so $A = K$. From step 3(i) of Proposition 1, $a_1/z_1 > K > a_2/z_2$, and therefore:

– $a_1 - Kz_1 > 0$, so $dU_1/dr > 0$: the rich loses welfare when r falls.

– $a_2 - Kz_2 < 0$, so $dU_2/dr < 0$: the poor gains welfare when r falls.

Hence the price effect is fully redistributive: $\Delta U_1 < 0 < \Delta U_2$.

A.6 Proof of Proposition 4, Item 1

To show that an increase in markups increases the equilibrium interest rate, we totally differentiate the equilibrium condition $A(r, \mu) = q(r, \mu)$ and obtain

$$dr^\mu = \frac{\varepsilon^{q,\mu} - \varepsilon^{A,\mu}}{\varepsilon^A + \varepsilon^q} d\mu > 0,$$

where $\varepsilon^{A,\mu} \equiv \partial \log A / \partial \mu < 0$ is the partial-equilibrium effect of markups on aggregate savings through the wage, and $\varepsilon^{q,\mu} \equiv \partial \log q / \partial \mu$ the partial-equilibrium effect of markups on the value of firms. This is positive as long as $\varepsilon^{q,\mu} > \varepsilon^{A,\mu}$, which we verify below.

We maintain $\beta = 0$ throughout, and evaluate locally at $\mu = 1$. The numerical values reported below refer to the two-agent economy with $\beta = 0$ at the parameter values of Section 4 of the main text ($\sigma = 2$, $\sigma_a = 0.89$, $\alpha = 1/3$, $\delta = 0.047$), with $z_1 = 0.7$ and Z set so that $r = 3\%$ at $\mu = 1$; $z_1 = 0.8$ gives the same signs.

Preliminaries: $dr/d\mu > 0$.

We first show that, in partial equilibrium (holding r constant), an increase in μ increases q . Write gross monopoly rents as $\Pi = \alpha \frac{\mu-1}{\mu} Y$. Then

$$\frac{\partial q}{\partial \mu} = \frac{\partial K}{\partial \mu} + \frac{1}{r} \frac{\partial \Pi}{\partial \mu}.$$

For the rent term, using the product rule:

$$\begin{aligned}
\frac{\partial \Pi}{\partial \mu} &= \alpha Y \cdot \frac{1}{\mu^2} + \alpha \frac{\mu - 1}{\mu} \cdot \frac{\partial Y}{\partial \mu} \\
&= \frac{\alpha Y}{\mu^2} - \alpha \frac{\mu - 1}{\mu} \cdot \frac{\alpha Y}{(1 - \alpha)\mu} \\
&= \frac{\alpha Y}{\mu^2} \left(1 - \frac{\alpha(\mu - 1)}{1 - \alpha} \right) = \frac{\alpha Y}{\mu^2} \cdot \frac{1 - \alpha\mu}{1 - \alpha}.
\end{aligned}$$

Combining and substituting $\alpha Y = \mu(r + \delta)K$:

$$\frac{\partial q}{\partial \mu} = \frac{K}{(1 - \alpha)\mu r} (\delta - \alpha\mu(r + \delta)) = \frac{K(r + \delta)}{(1 - \alpha)\mu r} \left(\frac{\delta}{r + \delta} - \alpha\mu \right).$$

Hence $\partial q / \partial \mu > 0$ if and only if $\mu < \delta / [\alpha(r + \delta)]$, which is satisfied at $\mu = 1$ for standard calibrations. Evaluating at $\mu = 1$, where $V = 0$ so that $q = K$:

$$\varepsilon^{q, \mu} \Big|_{\mu=1} \equiv \frac{1}{q} \frac{\partial q}{\partial \mu} \Big|_{\mu=1} = \frac{\delta}{r} - \frac{\alpha}{1 - \alpha}. \quad (11)$$

Effect of μ on savings supply. A change in μ also affects aggregate savings at fixed r through the wage. From the firm FOC, $\partial w / \partial \mu|_r = -\alpha w / [(1 - \alpha)\mu] < 0$. With $\beta = 0$, the steady-state Euler equation gives $c_i = a_i^\phi$ where $\phi \equiv \sigma_a / \sigma \in (0, 1)$. Combined with the budget constraint, $a_i^\phi - a_i r = w z_i$. Applying the IFT:

$$\frac{\partial a_i}{\partial \mu} \Big|_r = \frac{z_i}{D_i^*} \cdot \frac{\partial w}{\partial \mu} \Big|_r = -\frac{\alpha w z_i}{(1 - \alpha)\mu D_i^*},$$

where $D_i^* \equiv \phi a_i^{\phi-1} - r > 0$. Therefore

$$\varepsilon^{A, \mu} \equiv \frac{\partial \log A}{\partial \mu} \Big|_r = -\frac{\alpha w}{(1 - \alpha)\mu A} \sum_i \frac{z_i}{D_i^*} < 0. \quad (12)$$

At $\mu = 1$, using $\alpha w / [(1 - \alpha)A] = \alpha Y / A = (r + \delta)$:

$$|\varepsilon^{A, \mu}| \Big|_{\mu=1} = (r + \delta) \sum_i \frac{z_i}{D_i^*}. \quad (13)$$

Totally differentiating $A(r, \mu) = q(r, \mu)$:

$$\frac{dr}{d\mu} = \frac{\varepsilon^{q,\mu} - \varepsilon^{A,\mu}}{\varepsilon^A + \varepsilon^q} > 0, \quad (14)$$

since both $\varepsilon^{q,\mu} > 0$ and $-\varepsilon^{A,\mu} > 0$.

1. ε^q increases with μ .

A change in μ affects ε^q both directly (through the firm valuation) and indirectly (through the equilibrium interest rate):

$$\frac{d\varepsilon^q}{d\mu} = \underbrace{\frac{\partial \varepsilon^q}{\partial \mu} \Big|_r}_{\text{direct}} + \underbrace{\frac{\partial \varepsilon^q}{\partial r} \cdot \frac{dr}{d\mu}}_{\text{indirect}}. \quad (15)$$

Recall that $\varepsilon^q = \varepsilon^K + \delta(\mu - 1)/(r\Delta)$ where $\Delta \equiv \mu r + (\mu - 1)\delta$. Differentiating and evaluating at $\mu = 1$:

$$\frac{\partial \varepsilon^q}{\partial \mu} \Big|_r = \frac{\delta}{r^2} > 0, \quad \frac{\partial \varepsilon^q}{\partial r} = -\frac{\varepsilon^K}{r + \delta} < 0.$$

The direct effect is positive: higher markups raise firm valuations, increasing the sensitivity of asset supply to the interest rate. The indirect effect is negative: higher markups raise the equilibrium r , which reduces ε^q . Substituting $dr/d\mu$ from (14):

$$\frac{d\varepsilon^q}{d\mu} = \frac{\delta}{r^2} - \frac{\varepsilon^K}{r + \delta} \cdot \frac{\varepsilon^{q,\mu} - \varepsilon^{A,\mu}}{\varepsilon^K + \varepsilon^A}.$$

Since $\varepsilon^K + \varepsilon^A > \varepsilon^K$, the indirect term is bounded above (in absolute value) by its value at $\varepsilon^A = 0$:

$$\frac{d\varepsilon^q}{d\mu} > \frac{\delta}{r^2} - \frac{\varepsilon^{q,\mu} - \varepsilon^{A,\mu}}{r + \delta} = \frac{1}{r + \delta} \left(\frac{\delta^2}{r^2} + \frac{\alpha}{1 - \alpha} - |\varepsilon^{A,\mu}| \right). \quad (16)$$

This is positive provided

$$|\varepsilon^{A,\mu}| < \frac{\delta^2}{r^2} + \frac{\alpha}{1 - \alpha} \equiv \bar{\varepsilon}. \quad (17)$$

The threshold $\bar{\varepsilon}$ is of order $(\delta/r)^2$. Since $\sum_i z_i = 1$, $|\varepsilon^{A,\mu}| \leq (r + \delta)/\min_i D_i^*$, so condition (17) holds whenever $\min_i D_i^* > (r + \delta)/\bar{\varepsilon}$, a mild strengthening of the regularity

condition $D_i^* > 0$ of Assumption 1. At the calibrated values, $\bar{\varepsilon} \approx 3.0$, $(r + \delta)/\bar{\varepsilon} \approx 0.03$, $D_1^* = 0.036$ and $|\varepsilon^{A,\mu}| = 1.6$, so the condition holds; $d\varepsilon^q/d\mu = 24$, the sum of a direct effect of 52 and an effect of -28 through the equilibrium rise in r .

2. ε^A increases with μ .

The total change in ε^A along the equilibrium path decomposes as

$$\frac{d\varepsilon^A}{d\mu} = \underbrace{\frac{\partial \varepsilon^A}{\partial \mu} \Big|_r}_{\text{cross-partial}} + \underbrace{\frac{d^2 \log A}{dr^2} \cdot \frac{dr}{d\mu}}_{\text{convexity}}. \quad (18)$$

The convexity term is a product of two positive factors. We prove $d^2 \log A / dr^2 > 0$.

With $\beta = 0$, the budget equation is $a_i^\phi - a_i r = wz_i$. Define $D_i^* \equiv \phi a_i^{\phi-1} - r > 0$. The IFT gives $da_i/dr = \eta_i / D_i^*$, where $\eta_i \equiv a_i - Kz_i$. Differentiating once more (using $dD_i^*/dr = \phi(\phi-1)a_i^{\phi-2}a_i' - 1$ and $d\eta_i/dr = a_i' + K\varepsilon^K z_i$),

$$\frac{d^2 a_i}{dr^2} = \frac{2\eta_i}{(D_i^*)^2} + \frac{K\varepsilon^K z_i}{D_i^*} + \frac{\phi(1-\phi)a_i^{\phi-2}\eta_i^2}{(D_i^*)^3}. \quad (19)$$

The last term is non-negative because $\phi(1-\phi) > 0$: the concavity of the non-homothetic consumption function $c = a^\phi$ generates convexity in the savings response. At $\mu = 1$, $K = A$ so $\eta_1 + \eta_2 = 0$; write $\eta \equiv \eta_1 > 0$. Each summed component of $d^2 A / dr^2 = \sum_i d^2 a_i / dr^2$ is positive:

- (i) $2\eta[(D_1^*)^{-2} - (D_2^*)^{-2}] > 0$, since $a_1 > a_2$ and $\phi - 1 < 0$ imply $D_1^* < D_2^*$.
- (ii) $K\varepsilon^K[z_1/D_1^* + z_2/D_2^*] > 0$.
- (iii) $\phi(1-\phi)\eta^2[a_1^{\phi-2}/(D_1^*)^3 + a_2^{\phi-2}/(D_2^*)^3] > 0$.

It remains to show the stronger result $d^2 \log A / dr^2 > 0$, i.e. $A \cdot d^2 A / dr^2 > (dA/dr)^2$. Since $(dA/dr)^2 = \eta^2(D_2^* - D_1^*)^2 / [(D_1^*)^2(D_2^*)^2]$, term (i) alone suffices:

$$A \cdot (i) - (dA/dr)^2 = \frac{\eta(D_2^* - D_1^*)}{(D_1^*)^2(D_2^*)^2} [2A(D_1^* + D_2^*) - \eta(D_2^* - D_1^*)].$$

The bracket is positive: $2A > \eta$ (since $\eta = a_1 - Az_1 < Az_2 < A$) and $D_1^* + D_2^* > D_2^* - D_1^*$ (since $D_1^* > 0$). Terms (ii) and (iii) only add to the surplus. Therefore $d^2 \log A / dr^2 > 0$.

The cross-partial $\partial \varepsilon^A / \partial \mu|_r$ captures how the *slope* of the savings function shifts when markups change at fixed r . Its sign is ambiguous in general. However, $d\varepsilon^A / d\mu > 0$ whenever the convexity term dominates:

$$\left| \frac{\partial \varepsilon^A}{\partial \mu} \right|_r < \frac{d^2 \log A}{dr^2} \cdot \frac{dr}{d\mu}. \quad (20)$$

At the calibrated values, the cross-partial is -32 and the convexity term, the product of $d^2 \log A / dr^2 = 732$ and $dr / d\mu = 0.11$, is 80 , so condition (20) holds and $d\varepsilon^A / d\mu = 48$. Together with Step 1, $\varepsilon^A + \varepsilon^\eta$ rises with μ , from 24.2 at $\mu = 1$ to 28.4 at $\mu = 1.1$ and 30.2 at $\mu = 1.2$, and for a given $\partial \log A$ the equilibrium fall in r shrinks, which is item 1.

A.7 Proof of Proposition 4, Items 2 and 3

Part 1: $|(\varepsilon^{a_1} - \varepsilon^A)dr|$ **decreases with μ (holding $\varepsilon^{a_1} - \varepsilon^A$ constant).**

From the definition of the trickle-down effect,

$$|(\varepsilon^{a_1} - \varepsilon^A)dr| = \frac{(\varepsilon^{a_1} - \varepsilon^A) \partial \log A}{\varepsilon^A + \varepsilon^\eta}.$$

Since $\partial \log A$ is held fixed and $\varepsilon^A + \varepsilon^\eta$ increases with μ (item 1), $|(\varepsilon^{a_1} - \varepsilon^A)dr|$ decreases with μ to first order (holding $\varepsilon^{a_1} - \varepsilon^A$ constant).

Part 2: $\Omega \equiv \varepsilon^{a_1} / \varepsilon^A$ **decreases with μ (under $\beta = 0$, at $\mu = 1$).**

We maintain $\beta = 0$, $F = 0$ and evaluate locally at $\mu = 1$. Decompose the total derivative as

$$\frac{d\Omega}{d\mu} = \underbrace{\frac{\partial \Omega}{\partial \mu} \Big|_r}_{\text{direct}} + \underbrace{\frac{\partial \Omega}{\partial r} \Big|_\mu \cdot \frac{dr}{d\mu}}_{\text{indirect}}.$$

We show that the indirect effect is negative and dominates.

Step 1: Closed form. With $\beta = 0$, the Euler equation gives $c_i = a_i^\phi$ where $\phi \equiv \sigma_a/\sigma \in (0, 1)$, and the budget constraint reads $a_i^\phi - a_i r = w z_i$. Define $D_i^* \equiv \phi a_i^{\phi-1} - r > 0$. As in the proof of item 1, the total derivative of individual savings with respect to r is $da_i/dr = \eta_i/D_i^*$ where $\eta_i \equiv a_i - K z_i$. At $\mu = 1$, $K = A$ (since $V = 0$), so $\eta_1 + \eta_2 = 0$; write $\eta \equiv \eta_1 > 0$. The log semi-elasticities are

$$\varepsilon^{a_1} = \frac{\eta}{a_1 D_1^*}, \quad \varepsilon^A = \frac{\eta(D_2^* - D_1^*)}{A D_1^* D_2^*},$$

where $D_2^* > D_1^*$ since $a_1 > a_2$ and $\phi < 1$ imply $a_1^{\phi-1} < a_2^{\phi-1}$. Taking the ratio:

$$\Omega = \frac{A D_2^*}{a_1 (D_2^* - D_1^*)}. \quad (21)$$

Step 2: $d \log \Omega / dr < 0$ at fixed μ . Differentiating (21) logarithmically with respect to r :

$$\frac{d \log \Omega}{dr} = \underbrace{(\varepsilon^A - \varepsilon^{a_1})}_{< 0} + \underbrace{\frac{D_1^{*'} D_2^* - D_2^{*'} D_1^*}{D_2^* (D_2^* - D_1^*)}}_{< 0}, \quad (22)$$

where $D_i^{*'} \equiv dD_i^*/dr = \phi(\phi - 1)a_i^{\phi-2} \eta_i/D_i^* - 1$. The first term is negative because $\Omega > 1$ (Proposition 2). For the second, the cross-product evaluates to

$$D_1^{*'} D_2^* - D_2^{*'} D_1^* = \underbrace{\phi(\phi - 1)}_{< 0} \eta \left[\frac{a_1^{\phi-2} D_2^*}{D_1^*} + \frac{a_2^{\phi-2} D_1^*}{D_2^*} \right] - \underbrace{(D_2^* - D_1^*)}_{> 0}, \quad (23)$$

which is strictly negative since both terms have the same sign. Therefore $d \log \Omega / dr < 0$.

Step 3: Indirect effect dominates. Since $d \log \Omega / dr < 0$ and $dr/d\mu > 0$, the indirect effect reduces Ω . The direct effect $\partial \Omega / \partial \mu|_r$ works through the wage at a fixed r and has no definite sign, so item 3 holds whenever

$$\left. \frac{\partial \Omega}{\partial \mu} \right|_r < - \left. \frac{\partial \Omega}{\partial r} \right|_\mu \cdot \frac{dr}{d\mu}. \quad (24)$$

At the calibrated values, the direct effect is 1.9 and the indirect effect -5.1 , so $d\Omega/d\mu =$

–3.2: Ω falls from 1.34 at $\mu = 1$ to 1.20 at $\mu = 1.1$ and 1.16 at $\mu = 1.2$.

Implication for the trickle-down effect. The trickle-down effect can be decomposed as

$$|(\varepsilon^{a_1} - \varepsilon^A)dr| = \underbrace{(\Omega - 1)}_{\text{excess sensitivity}} \cdot \underbrace{\frac{\varepsilon^A}{\varepsilon^A + \varepsilon^q}}_{\equiv s} \cdot \partial \log A,$$

where s denotes the demand share of asset-market clearing. As μ increases, $\Omega - 1$ falls (item 3) and $\varepsilon^A + \varepsilon^q$ rises (item 1), while the rise in ε^A itself works in the other direction, so the net effect is not signed in general. At the calibrated values, $(\Omega - 1)s$ falls from 0.066 at $\mu = 1$ to 0.059 at $\mu = 1.1$ and 0.057 at $\mu = 1.2$: for a given $\partial \log A$, the trickle-down effect weakens with the markup.

B Additional Details on Calibration

B.1 Deriving the Pareto tail in a Heterogeneous-agent model

The proof draws heavily on Gaillard et al. (2023). We derive the asymptotic right tails of the wealth and the consumption distributions in a simpler framework without scale and type dependence. All variables are detrended by $(1 + g)^t$, $\tilde{\beta}$ is the detrended discount factor, and we write $R \equiv (1 + r)/(1 + g)$ for the gross return on detrended wealth.

The problem of the household writes

$$V(a_t, z_t) = \frac{c_t^{1-\sigma}}{1-\sigma} + \gamma \frac{(a_{t+1} + \zeta)^{1-\sigma_a}}{1-\sigma_a} + \tilde{\beta}(1 - \xi)\mathbb{E}[V(a_{t+1}, z_{t+1})]$$

subject to

$$a_{t+1} + c_t = R a_t + z_t w$$

and a zero borrowing constraint

$$a_{t+1} \geq 0.$$

Households are subject to persistent idiosyncratic productivity shocks z_t and to random death shocks arriving with probability ξ . The Euler equation of the household is

$$c_t^{-\sigma} = (1 - \xi)\tilde{\beta}R \mathbb{E}[c_{t+1}^{-\sigma}] + \gamma (a_{t+1} + \zeta)^{-\sigma_a}.$$

The key thing to note is that, when $a \rightarrow \infty$, labor income $z_t w$ becomes negligible and we can abstract from uncertainty coming from idiosyncratic productivity shocks. The problem now writes

$$V(a_t) = \frac{c_t^{1-\sigma}}{1-\sigma} + \gamma \frac{(a_{t+1} + \zeta)^{1-\sigma_a}}{1-\sigma_a} + \tilde{\beta}(1 - \xi)V(a_{t+1})$$

subject to

$$c_t + a_{t+1} = R a_t.$$

The Euler equation now writes

$$c_t^{-\sigma} = (1 - \xi)\tilde{\beta}R c_{t+1}^{-\sigma} + \gamma (a_{t+1} + \zeta)^{-\sigma_a}.$$

where $c_t + a_{t+1} = R a_t$. Guess that $c = B a^{\sigma_a/\sigma}$, this implies that

$$a' = R a - c = R a - B a^{\sigma_a/\sigma}.$$

This implies that as $a \rightarrow \infty$, $a' \sim R a$: detrended wealth of the rich grows at $(1 + r)/(1 + g)$. We use a guess and verify technique. As $a \rightarrow \infty$, we have

$$c' = B(a')^{\sigma_a/\sigma} = B(R a - B a^{\sigma_a/\sigma})^{\sigma_a/\sigma} = B(R a)^{\sigma_a/\sigma}$$

$$c^{-\sigma} = (1 - \xi)\tilde{\beta}R B^{-\sigma}(R a)^{-\sigma_a} + \gamma(R a)^{-\sigma_a},$$

using $a' \sim R a$ in the taste-for-wealth term as well, so that

$$c = [R a]^{\sigma_a/\sigma} \left[(1 - \xi)\tilde{\beta}R B^{-\sigma} + \gamma \right]^{-1/\sigma}.$$

Which verifies our guess. To find B , solve

$$B = R^{\sigma_a/\sigma} \left[(1 - \xi) \tilde{\beta} R B^{-\sigma} + \gamma \right]^{-1/\sigma}$$

with a numerical solver. A surviving household therefore has $a' \sim R a$, while a dying household's wealth is reset to the bequest endowment, which is negligible in the tail.

We can now derive the Pareto tail in this case. Beare & Toda (2022) show that the stationary wealth distribution has a Pareto upper tail, characterized by the moment generating function

$$M_{ss'}(z) = \mathbb{E}[e^{z \log G'} | s_t = s, s_{t+1} = s'] = \mathbb{E}[G'^z | s_t = s, s_{t+1} = s'].$$

In our case, define $s = 0$ an alive household and $s = 1$ a dead household. We have

$$G_{s=0} = R, \quad G_{s=1} = 0$$

The matrix $M(z) = (M_{ss'}(z))$ thus writes

$$M(z) = \begin{bmatrix} R^z & 0 \\ R^z & 0 \end{bmatrix}$$

And the Pareto tail exponent θ^a is then defined by

$$\rho(P \circ M(z)) = 1$$

where $\circ$ is the Hadamard (entrywise) product and ρ the spectral radius. We have

$$P \circ M(z) = \begin{bmatrix} (1 - \xi) R^z & 0 \\ (1 - \xi) R^z & 0 \end{bmatrix}$$

And we have $\rho(P \circ M(z)) = (1 - \xi)R^z = 1$. Solving for z yields

$$z = \theta^a = \frac{-\log(1 - \xi)}{\log R} = \frac{-\log(1 - \xi)}{\log \left(\frac{1+r}{1+g} \right)}$$

which is the Pareto tail. Without growth ($g = 0$) the drift is $1 + r$ and the same tail requires a much higher death rate.

To obtain the Pareto tail for consumption, note that $c = Ba^{\sigma_a/\sigma}$ implies that $a = c^{\sigma/\sigma_a} B^{-\sigma/\sigma_a}$. We know that a follows a Pareto distribution with tail θ^a . Thus,

$$P(c > x) = P(Ba^{\sigma_a/\sigma} > x) = P\left(a > \frac{x^{\sigma/\sigma_a}}{B}\right)$$

Since we know that $P(a > y) \sim Cy^{-\theta^a}$, we have $P(c > y) \sim C((y/B)^{\sigma/\sigma_a})^{-\theta^a}$. This means that the distribution for consumption follows a Pareto distribution with tail

$$\theta^c = \theta^a \sigma / \sigma_a.$$

Now, Gaillard et al. (2023) show that in the U.S., $\theta^c / \theta^a \approx 2.24$. Setting $\sigma = 2$, this implies $\sigma_a = 0.89$.

With $r = 3\%$, $g = 1.5\%$ and a Pareto tail for wealth of $\theta^a = 1.35$, this yields a probability to die of

$$\xi = 1 - R^{-\theta^a} = 1 - \left(\frac{1+r}{1+g} \right)^{-\theta^a} \approx 0.020,$$

against $1 - (1 + r)^{-\theta^a} \approx 0.04$ without growth.

B.2 The 1970 wealth distribution in the model and in the data

Table B.1 reports the 1970 distribution of wealth in the model and in the data.¹ The model matches the top shares, where one-asset models with precautionary saving alone fall short: the taste for wealth makes the rich save almost all of their income (their policy

¹Online Appendix D describes the numerical solution method.

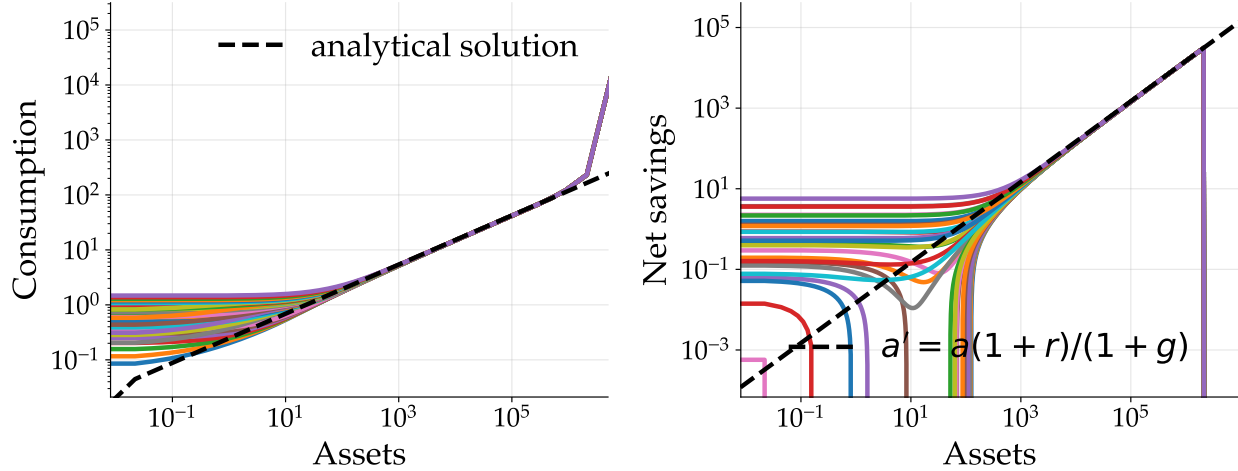


Figure B.1: Policy function for savings and consumption at the steady state

Note: The first panel shows the consumption function, as a function of wealth (x-axis), and idiosyncratic types (different colors), versus our analytical solutions for high levels of wealth. The second panel plots the policy function for the change in wealth ($a' - a$) versus the approximation $a' = a(1+r)/(1+g)$.

function converges to $a' = a(1+r)/(1+g)$, Figure B.1), and the death probability keeps the distribution from degenerating.

Table B.1: Distribution of wealth in the baseline model for the year 1970

Name	Bottom 50%	Next 40%	Top 10%	Top 1%	Top 0.1%	Top 0.01%	r	A/Y
Model	6.6	33.0	60.4	26.9	11.0	4.4	3.0	3.7
Data PSZ	1.0	29.9	69.1	26.9	9.5	3.3		

Note: This table shows the distribution of wealth in the baseline model, along with the equilibrium interest rate and wealth-to-output ratio, in levels. For example, the 'Top 10%' column of the model row is the top 10% wealth share in percent.

B.3 Contribution of each ingredient to the 1970 wealth distribution

Table B.2 removes one ingredient of the model at a time, keeping the calibration otherwise unchanged.² The taste for wealth is what generates the concentration: without it, the top 1% share falls by 19.1 percentage points and the top 0.1% share by 9.7. Removing imperfect competition at fixed technology lowers the top 1% and top 0.1% shares by

²Without the taste for wealth, we re-solve β to keep r at 3% and the wealth-to-output ratio at its baseline value; keeping β at its calibrated value would imply an interest rate near 100%. In the other rows β stays at its baseline value and r adjusts to clear the asset market. Because the model is non-linear, the rows do not add up to the baseline.

about 5 percentage points, because the economy then holds its wealth as physical capital, which lowers the interest rate by 0.8 percentage points and raises wages; the fall in r hits the wealthy, whose income is mostly financial, while the rise in w benefits the bottom of the distribution. Paying bequests equally rather than in proportion to permanent income barely moves the top shares. Removing idiosyncratic productivity shocks raises the top 1% share by 31 percentage points: as in Hubmer et al. (2021), these shocks generate precautionary saving at the bottom of the distribution and smooth the wealth distribution, so their variance lowers rather than raises wealth inequality.

Table B.2: Contribution of different channels to steady state wealth shares

#	Scenario	Bottom 50%	Next 40%	Top 10%	Top 1%	Top 0.1%	Top 0.01%	r	A/Y
1	Homothetic prefs. ($\gamma = 0$)	8.4	14.2	-22.7	-19.1	-9.7	-4.2	0.0	0.0
2	Perfect competition ($\mu = 1$)	-0.9	1.7	-0.9	-4.8	-4.8	-3.2	-0.8	-120.9
3	No prop. inheritance	2.8	-1.0	-1.8	0.1	0.7	0.7	0.1	-9.1
4	No trans. shocks ($\sigma_\varepsilon = 0$)	-6.5	-20.9	27.3	30.6	26.2	20.3	0.5	-50.7

Note: This table displays the p.p. change in the wealth distribution, the interest rate, and the wealth-to-output ratio, for each quantile after removing a given feature of the model, keeping the calibration constant. For example, the first row reports the effect of removing the non-homothetic taste for wealth ($\gamma = 0$): the top 1% wealth share changes by -19.1 p.p. and the interest rate by 0.0 p.p. (from 3.0% to 3.0%).

B.4 Marginal propensities to save by permanent income type

Table B.3 reports the marginal propensity to save of each permanent income type in the baseline steady state: the additional wealth accumulated over six years after a small permanent rise in post-tax labor income, divided by the cumulated additional income, at fixed prices.

Table B.3: Marginal propensity to save by permanent income group

Bottom 50%	Next 40%	Next 9%	Next 0.9%	Top 0.1%
0.12	0.32	0.59	0.81	0.86

Note: This table shows the marginal propensity to save (MPS) for each permanent income group at the initial steady state: the additional wealth accumulated over 6 years after a small permanent rise in post-tax labor income, divided by the cumulated additional income (prices held fixed).

B.5 Unit of observation: equal-split adjustment of the SCF

The Distributional National Accounts of Piketty et al. (2018) report inequality statistics per *equal-split adult*: for married couples, total household income and wealth are divided equally between the two spouses, and each spouse is treated as a separate observation.

The Survey of Consumer Finances (SCF), by contrast, uses the household—or more precisely, the *primary economic unit* (PEU)—as its unit of observation. All income and wealth variables are reported at the household level, without distinguishing between spouses.

To ensure comparability between the two data sources, we convert the SCF to a per-adult basis by applying an equal-split adjustment: for each married or partnered household (identified by the `MARRIED` variable in the SCF), we divide all dollar-denominated variables by two and multiply the survey weight by two. Single-person households are left unchanged. This transformation ensures that top shares and portfolio compositions computed from the SCF are measured on the same per-adult basis as the PSZ distributional data and as the agents in our model.

B.6 Calibration of the exogenous portfolio share

We follow Auclert & Rognlie (2018) and calibrate our portfolio share function on data from the Survey of Consumer Finances, using all available waves from 1989 to 2019, converted to a per-adult basis. The narrow share, used in the baseline, is the share of net worth held in claims on firms, with the same definition as the risky assets of Figure 4:

$$\text{narrow share} = \frac{\text{EQUITY} + \text{BUS} + \text{OTHNFIN}}{\text{NETWORTH}},$$

where `EQUITY` is equity held directly and indirectly through mutual funds, retirement accounts, annuities and trusts, `BUS` is private business equity, and `OTHNFIN` other non-financial assets. The broad share, used in the robustness analysis, adds housing and other

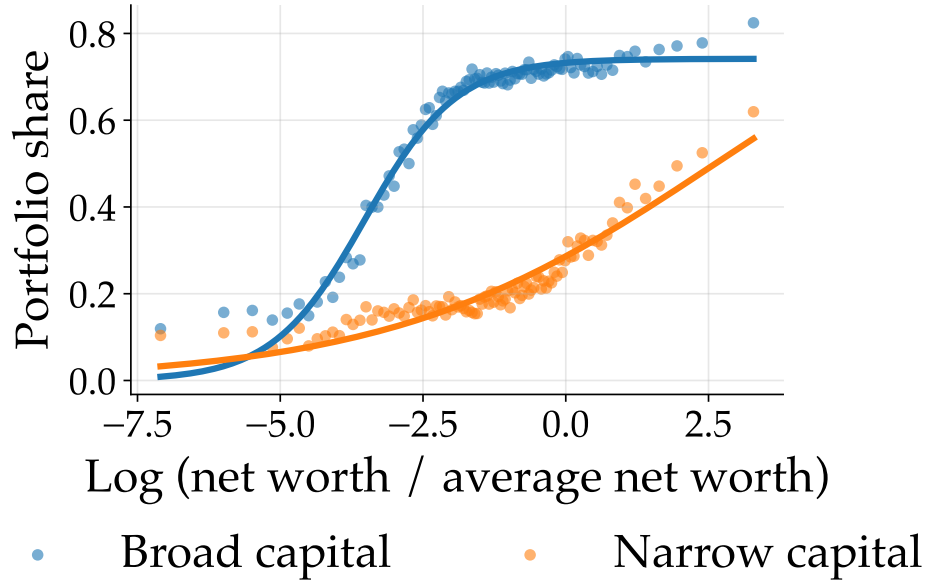


Figure B.2: Portfolio shares in the SCF and in the model

real estate net of the debt secured on it:

$$\text{broad share} = \frac{\text{EQUITY} + \text{BUS} + \text{OTHNFN} + \text{net housing}}{\text{NETWORTH}},$$

$$\text{net housing} = \text{HOUSES} + \text{ORESRE} + \text{NNRESRE} - \text{MRTHL} - \text{RESDBT}.$$

We compute weighted averages of each share by centile of net worth relative to the wave mean and fit the logistic function of Equation (3) on these centile averages, with the level parameter bounded by one since the model caps the share at one.

Figure B.3 and Table B.4 document the composition of wealth by wealth group in the distributional national accounts of Piketty et al. (2018) and in the SCF: equity and business wealth dominate at the top, housing in the bottom 90%, and unrealized capital gains are a larger share of net worth at the top.

Table B.5 reports the share of net worth, of claims on firms and of claims on firms plus net housing held by each wealth group in the SCF: the bottom half of the distribution owns almost none of either risky aggregate, and the two definitions differ in how they split the rest between the next 40% and the top 1%.

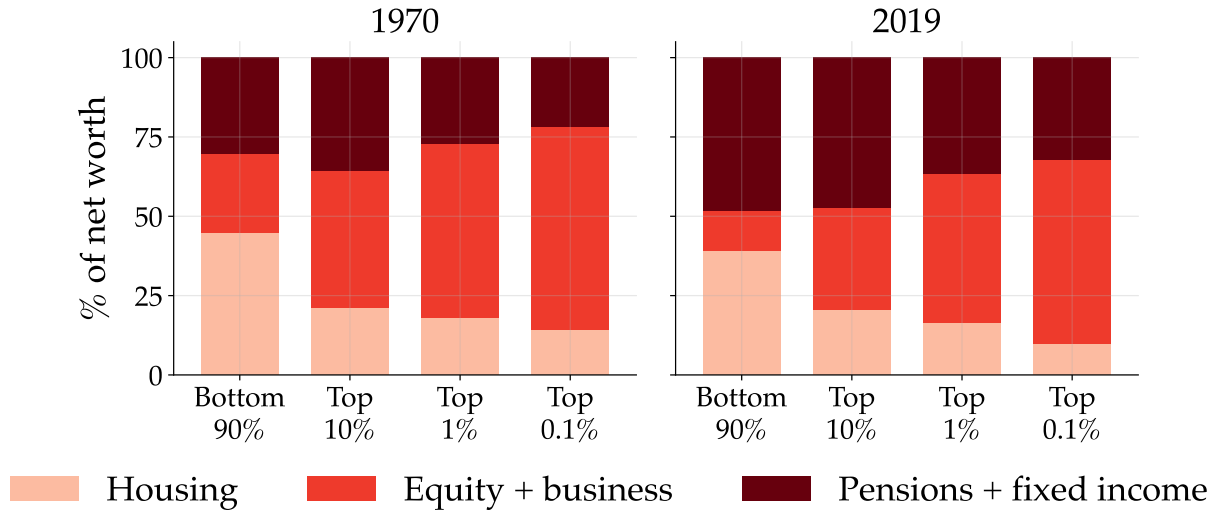


Figure B.3: Composition of net worth by wealth group, 1970 and 2019

Note: Distributional national accounts (Piketty et al. 2018), equal-split adults. Housing is net of mortgages; fixed-income claims are net of non-mortgage debt.

B.7 Markups over Time in De Loecker et al. (2020)

In the series of De Loecker et al. (2020), the average markup between 1970 and 2017 is 1.39, and the average between 1960 and 1980 is 1.3. Our baseline of 1.2 is at the lower end of the 1960s and 1970s values; the main text reports results for μ between 1.0 and 1.3.

Table B.4: Portfolio composition by wealth group

	DINA 1970		DINA 2019		SCF 1989-2019		
	Housing	Equity+business	Housing	Equity+business	Housing	Equity+business	Capital gains
Bottom 90%	44.8	24.8	39.1	12.6	53.4	28.1	32.2
Top 10%	21.2	43.1	20.6	32.0	25.3	55.3	38.0
Top 1%	18.0	54.8	16.5	46.9	18.2	66.0	45.6
Top 0.1%	14.3	63.8	10.0	57.8	13.9	73.1	54.1

Note: Percent of the group's net worth. DINA: Piketty, Saez and Zucman (2018, update 2023), Appendix Tables E2, E2b, E2c, equal-split adults; housing is net of mortgages, equity + business = equities + business assets; the remainder is pensions and fixed-income claims net of non-mortgage debt. SCF: equal-split adults, means over the 1989-2019 waves; housing is net of mortgages, equity + business = BUS + EQUITY + OTHNFIN, capital gains = unrealized capital gains (KGTOTAL) in percent of net worth; bottom 90% = all households minus the top 10%.

Table B.5: Ownership of net worth and of risky assets by wealth group

	Net worth	Firm claims (narrow)	Firm claims and net housing (broad)
Bottom 50%	4.6	1.7	4.0
Next 40%	27.7	15.9	25.8
Top 10%	67.7	82.3	70.1
Top 1%	31.3	45.7	34.2
Top 0.1%	11.4	18.3	12.8

Note: Share of the aggregate held by each wealth group, in percent, averaged over the 1989–2019 waves of the SCF (equal-split adults, sample of the portfolio fit of Online Appendix B.6). Firm claims are BUS + EQUITY + OTHNFIN; net housing is HOUSES + ORESRE + NNRESRE - MRTHEL - RESDBT. Groups are ranked by net worth within each wave.

C Additional Quantitative Results

This section reports additional quantitative results. Table C.1 reports the long-run changes in prices and quantities between the 1970 and 2019 steady states, and Table C.2 the steady-state general-equilibrium effect on wealth shares, the counterpart of Table 3 in the main text: the decline in r lowers the top shares and the bequest flow raises the bottom shares, and the total effect lowers the top 1% share by 6.8 points under perfect competition and by 6.5 points with markups.

Figure C.1 decomposes the general-equilibrium effect on the average wealth of each group, channel by channel, after a one-time shock, and Figure C.2 shows the composition of income along the wealth distribution in the first period of the shock, in the baseline model.

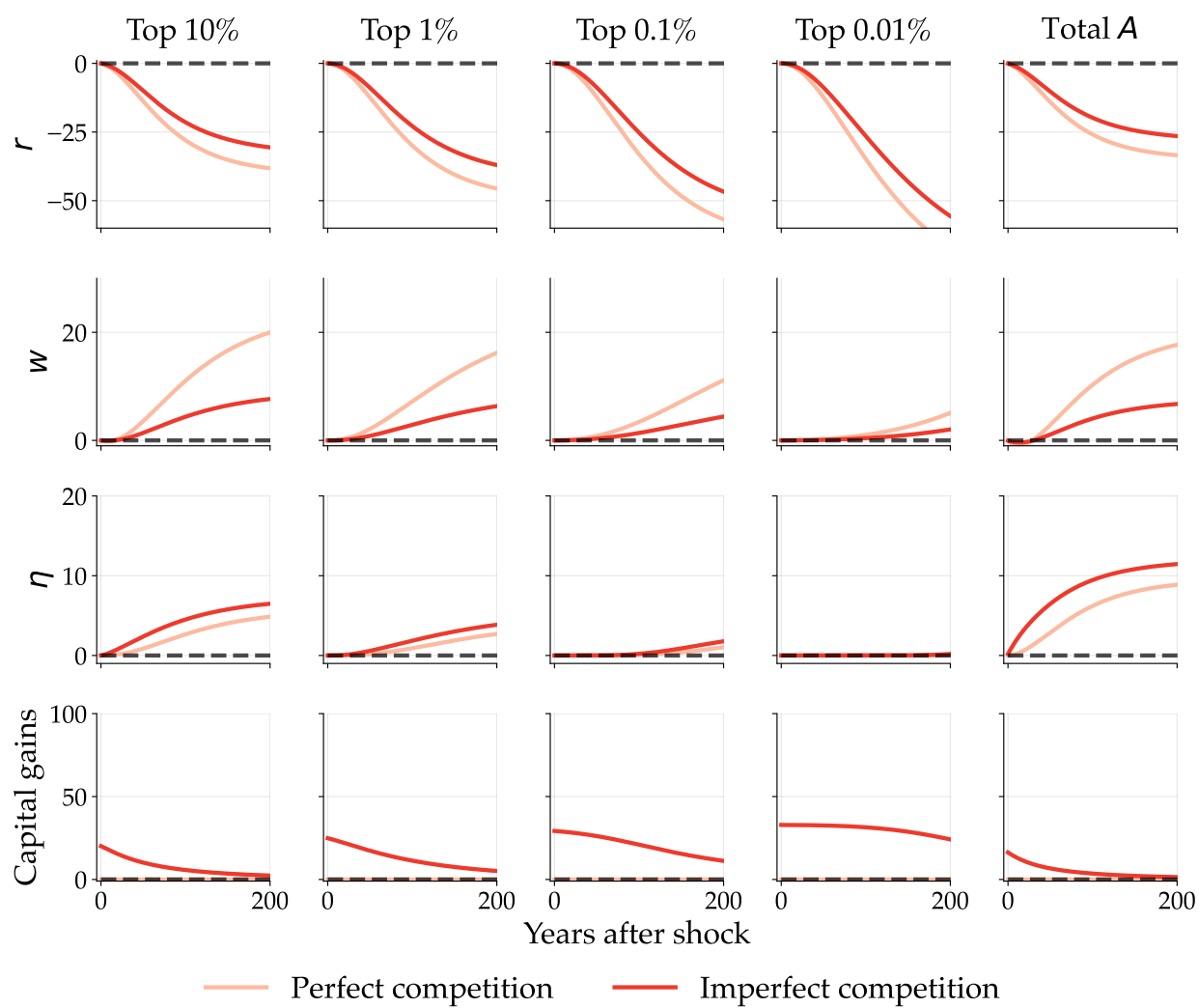


Figure C.1: Decomposition of the GE effect on the per-quantile average wealth levels.

Table C.1: Change in macro variables between initial and terminal steady states

Variable	$\mu = 1.0$	$\mu = 1.2$
r	-0.8	-0.6
w	9.6	3.8
η	29.0	36.1
Y	9.6	3.8
K	31.6	11.9
q	31.6	38.9
G	2.9	4.1

Note: This table shows the change in key macro variables between the initial (1970) and terminal steady states. The interest rate r is reported as a percentage point (p.p.) change. All other variables are reported as percentage (%) changes.

Table C.2: General equilibrium effect on wealth shares at the steady state

Variable	Model	Bottom 50%	Next 40%	Top 10%	Top 1%	Top 0.1%	Top 0.01%
r	$\mu = 1.0$	0.4	4.7	-5.1	-6.1	-5.1	-3.2
	$\mu = 1.2$	0.4	3.8	-4.3	-5.2	-4.4	-2.9
w	$\mu = 1.0$	-0.7	-1.3	2.0	0.9	0.4	0.3
	$\mu = 1.2$	-0.3	-0.6	0.9	0.4	0.2	0.1
η	$\mu = 1.0$	0.6	1.5	-2.1	-1.3	-0.6	-0.2
	$\mu = 1.2$	0.7	1.8	-2.5	-1.6	-0.7	-0.3
Total	$\mu = 1.0$	0.4	5.1	-5.5	-6.8	-5.4	-3.3
	$\mu = 1.2$	0.9	5.3	-6.1	-6.5	-4.9	-3.0

Note: This table displays the p.p. change in the wealth shares for each quantile due to the general equilibrium effect. Note that due to non-linearities in the policy functions, the sum of the partial effects does not add up to the total general equilibrium effect.

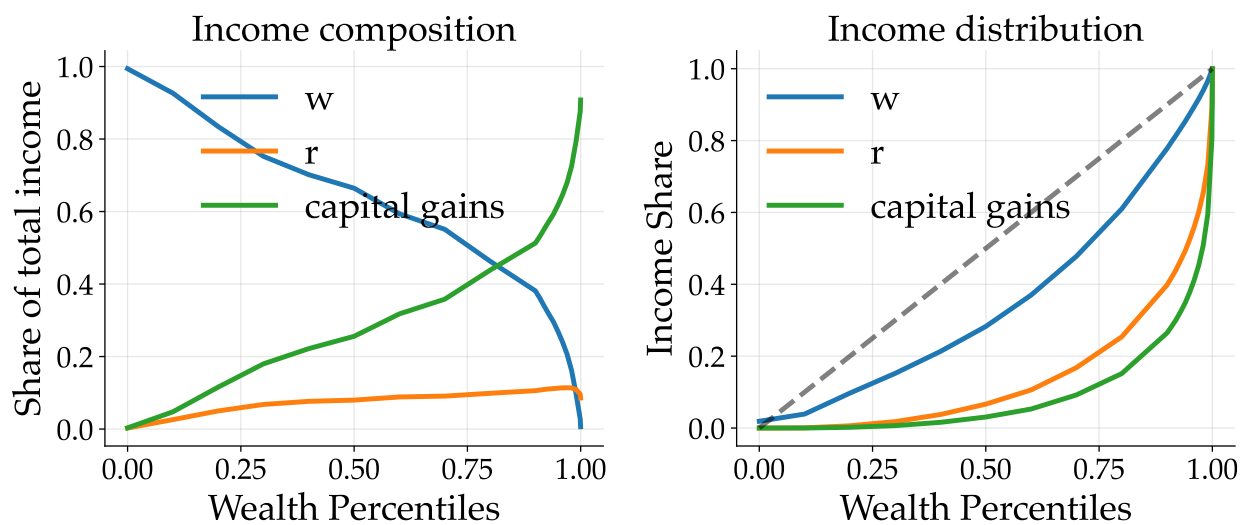


Figure C.2: Income composition and distribution along the distribution of wealth, at period $t = 0$ of a one-time permanent labor income inequality shock

C.1 Gradual Transition: Additional Results

Table C.3 and Figure C.3 compare the top wealth shares, the wealth-to-output ratio and Tobin’s Q along the gradual (myopic) transition with U.S. data over 1970 to 2020; the price paths are in Section 6 of the main text. Table C.4 reports the robustness table of the main text for all six wealth groups.

Table C.3: Model and data, 1970 to 2020

	Data	Model	Price channel	Model, $\mu = 1$
Top 10% wealth share (pp)	2.35	8.05	1.28	7.43
Top 1% wealth share (pp) [†]	8.01	8.46	2.23	7.08
Top 0.1% wealth share (pp) [†]	8.11	3.43	1.65	2.28
Top 0.01% wealth share (pp)	5.86	0.58	0.88	-0.16
Wealth-to-output W/Y (pp) [†]	208.65	83.56	31.02	21.30
Tobin’s Q (level)	0.73	0.20	0.20	0.00
Capital-gains share of Δ wealth (%)	74.30	72.87	–	0.00

Note: Changes from 1970 to 2019 (data, PSZ) and from 1970 to 2020 (model, gradual transition with annual surprises). Model: full transition at $\mu = 1.2$; the price channel holds permanent labor income at its 1970 distribution and feeds households the realized paths of r , w , capital gains and bequests; the last column is the full transition under perfect competition. [†]: the 1970 level of this moment is a calibration target (the top 0.1% share through the targeted Pareto tail); no change over time is targeted. The capital-gains share of accumulation is the sum of capital gains over 1971–2019 divided by the change in wealth: residual capital gains in the data (PSZ), unexpected equity revaluations in the model; Tobin’s Q is an aggregate price, so its price-channel cell repeats the total. The table quantifies one mechanism: the transition includes no force other than rising permanent labor income inequality.

C.2 Robustness Variants

Table 4 in the main text reports the price (general-equilibrium) channel effect on the average wealth level of each group in 2020, in percent of its 1970 steady-state level: the change in wealth implied by the realized paths of r , w , asset prices and bequests, holding the distribution of permanent labor income at its 1970 value. Every variant is a gradual transition fed with the same 1970–2019 path of permanent labor income inequality, and differs from the baseline in one respect.

Markup. The markup variants use annual surprises and the baseline portfolio rule. At $\mu = 1$, Z and δ are re-pinned so that r , w and the wealth-to-output ratio are the same as in

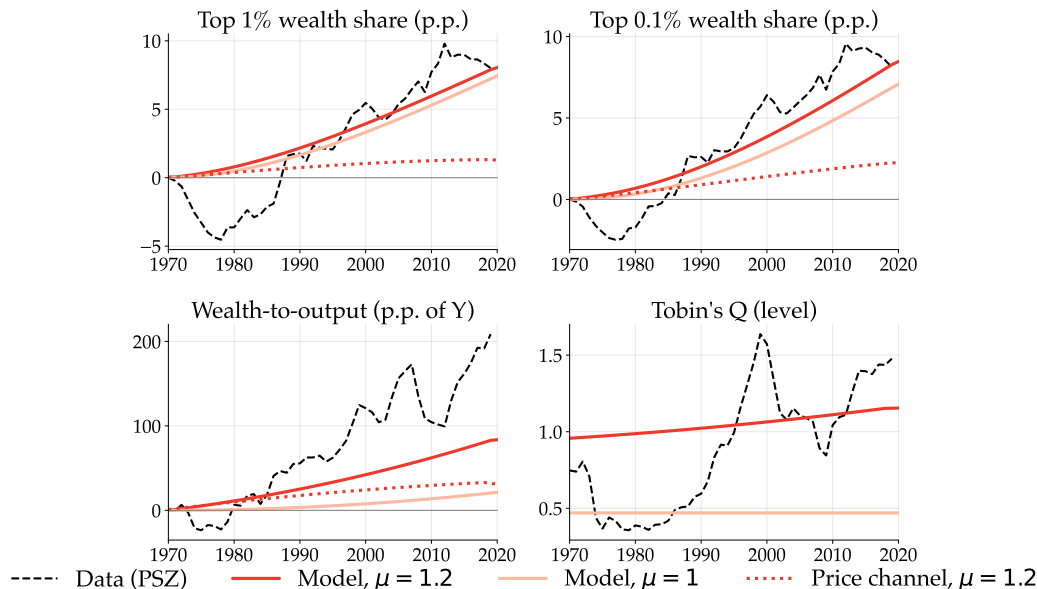


Figure C.3: Model and data over time, 1970 to 2020

Note: Top 1% and top 0.1% wealth shares and the household wealth-to-output ratio as changes since 1970 (percentage points), and Tobin's Q in levels. Data: Piketty et al. (2018), 1970 to 2019. Model: gradual transition with annual surprises, total (solid) and price channel alone (dotted), $\mu = 1.2$ and $\mu = 1$. The price channel has no separate Tobin's Q.

the baseline, which leaves the household side, and therefore the 1970 wealth distribution, unchanged; only the asset-supply side differs. At $\mu = 1.1$ and $\mu = 1.3$, technology is kept at its baseline value and β is re-solved to clear the asset market at $r = 3\%$ (Section 4 of the main text).

Portfolio rule. The baseline allocates the capital gains of each revelation date with the narrow rule fitted to the share of net worth held in claims on firms in the pooled 1989–2019 SCF (Online Appendix B.6). The broad rule adds housing net of mortgages to these claims; the constant-share rule gives every household the same exposure, so that capital gains are allocated in proportion to wealth.

Expectations. The baseline is a sequence of annual surprises: each year, households learn the current change in the income distribution and expect no further change. In the window variants, households learn the next 5 or 10 years of the inequality path at once and re-optimize when the next block is revealed. Under perfect foresight, the whole 1970–2019 path is revealed in 1970, so the revaluation of firms happens once, on impact.

Growth, bequests and taxes. The no-growth variant sets $g = 0$, recalibrates the death rate

Table C.4: Price channel effect on average wealth levels in 2020 across model variants, all wealth groups

	Bottom 50%	Next 40%	Top 10%	Top 1%	Top 0.1%	Top 0.01%	Average
<i>Markup</i>							
$\mu = 1.0$	-8.9	-5.2	-4.6	-4.0	-4.1	-4.3	-5.4
$\mu = 1.1$	0.6	3.2	8.7	15.0	21.9	27.1	6.3
$\mu = 1.2$ (baseline)	3.8	5.3	10.8	17.6	24.8	30.3	8.4
$\mu = 1.3$	4.9	6.0	11.3	17.9	24.8	30.1	8.9
<i>Portfolio rule ($\mu = 1.2$)</i>							
Broad incl. housing	6.6	7.1	8.4	10.9	12.7	13.2	7.8
Constant share	7.0	7.1	8.3	10.7	12.4	12.9	7.8
<i>Expectations ($\mu = 1.2$)</i>							
5-year window	3.9	5.1	10.0	16.4	23.4	28.8	7.9
10-year window	4.1	4.9	9.1	15.2	21.9	26.9	7.4
Perfect foresight	5.7	4.8	5.6	9.6	14.5	18.2	5.3
<i>Growth, bequests, taxes ($\mu = 1.2$)</i>							
No growth ($g = 0$)	-4.0	-3.0	-1.4	0.1	1.1	1.9	-2.3
Equal bequest redistribution	3.3	4.2	8.4	14.9	21.2	26.1	6.4
Flat labor tax ($\lambda = 0$)	6.4	8.2	15.0	23.3	32.9	40.3	12.4

Note: Price (general equilibrium) channel effect on the average wealth level of each group in 2020, in percent of its 1970 level, holding the income distribution at its 1970 value, for all six wealth groups; the variants are those of Table 4 in the main text.

to the wealth Pareto tail, and re-pins technology to the same 1970 targets for r , w and the wealth-to-output ratio, so that rents are capitalized at r instead of $r - g$. The equal-bequest variant splits the net bequest pool equally across newborn households instead of in proportion to permanent income. The flat-tax variant sets the progressivity of the labor income tax to zero, so that the shock fed to the model is the pre-tax change in permanent income shares rather than the post-tax change, which is smaller by a factor $1 - \lambda$.

D Solution Method

Household problem. We solve the household's dynamic programming problem using the endogenous grid method (EGM) of Carroll (2006). Because the wealth distribution exhibits a Pareto tail, we discretize the asset space on a non-linear grid of $n_A = 1,000$ points with curvature chosen to concentrate points in the empirically relevant range while extending the upper bound to $\bar{a} = 10^7$ (about 2.7 million times average wealth). We verify

ex-post that the mass of the stationary distribution at the upper bound is negligible (below 10^{-8}), ensuring that the grid does not artificially truncate the tail.

Steady state. At the steady state, the equity price follows from the no-arbitrage condition of the main text, $q = (1 + g)(1 - \tau_k)d/(r - g)$, reducing the general equilibrium problem to a single unknown. To calibrate the initial steady-state, we solve for the discount factor β such that the asset market clears: $A = q + B$. For the terminal steady-state (with high-inequality), we find the level of physical capital K that clears the asset market. The outer loop uses Brent’s method, while each evaluation solves the full household problem via EGM.

One-time transition. We compute perfect-foresight transitions following an unexpected, permanent MIT shock to the distribution of permanent labor income. Given initial and terminal steady states, we solve for the T -period path of the capital stock $\{K_t\}_{t=0}^{T-1}$ that clears the asset market at every date. We initialize the Jacobian of the asset market clearing condition with respect to K using the sequence-space Jacobian of Auclert et al. (2021), and then update it using Broyden’s method. At each iteration, given a candidate $\{K_t\}$, we compute the implied price sequences $\{r_t, w_t, q_t\}$ from the firm and arbitrage blocks, solve the household problem forward from the initial distribution, and evaluate the asset market residual. The Broyden update avoids recomputing the full $T \times T$ Jacobian at each step, making the algorithm tractable for long horizons ($T = 500$).

Myopic transition. The myopic transition models the gradual increase in permanent labor income inequality as a sequence of unanticipated MIT shocks. At each date $t \in \{1, \dots, T_{\text{trans}}\}$, the economy is hit by a new shock to the permanent income distribution $\{z_i\}_t$, and households treat it as one-time and permanent. Conditional on the shock, they have perfect foresight over future prices. Formally, at date t we solve a full perfect-foresight transition from the current state of the economy to the new terminal steady state implied by $\{z_i\}_t$. We then extract the period- t outcomes—prices, distributions, and aggregates—and use the resulting distribution Φ_{t+1} as the initial condition for the next

shock at $t + 1$. This nests a sequence of MIT-shock transitions, each solved using the Broyden algorithm described above.

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